SkillUp

Final Project: AI-Powered Visualizations

Estimated time: 60 minutes

Learning objectives

After completing the final project, you will be able to:

- Use generative AI tools to create a basic dashboard with given data
- Use generative AI tools to create a more complex dashboard with given data, using prompting
- Prepare a business case for your dashboards and visualizations created using generative AI tools
- Design and compile a comprehensive data presentation using generative AI tools, showcasing dynamic dashboards and visualizations

Introduction

Generative AI tools for Business Intelligence (BI) analysis is a tool to automate the creation of insightful and visually appealing data representations. It enables the swift generation of various visualizations, such as bar charts, line charts, and scatter plots, tailored to key metrics and trends. Analysts can use AI-powered tools to implement advanced techniques like predictive analytics and machine learning algorithms, uncovering hidden patterns and insights in the data. These tools can also generate interactive dashboards that allow users to explore the data and gain deeper insights, enhancing the overall analysis. The automation provided by these AI tools significantly streamlines data preparation and analysis, allowing analysts to focus on interpreting the results and making strategic decisions. This capability not only enhances the efficiency of the BI process but also democratizes data analysis, making it more accessible to non-technical stakeholders, thereby fostering a data-driven decision-making culture across the organization.

Note: Please use the latest version of your browser for this project. You can consider shifting the prompts based on your organization's requirements.

Also, remind you use the same prompt given in the project for practice, you may receive different responses.

Exercise: AI-driven sales analysis

For this project, you will work with Superstore's sample sales dataset. SkillUp's AI-powered BI tool designed to assist both consumers and businesses.

The dataset includes detailed sales data, orders, and images. ClaidAI's AI-powered BI tool leverages these datasets to create insightful visualizations, automate repetitive tasks, and gain deeper insights through AI-driven analysis.

Scenario

You are tasked with the following scenario for this exercise:

As a new data & Business Intelligence (BI) analyst, you are responsible for smart sales. Recently, the company launched a new initiative to improve sales performance and customer satisfaction. To support this initiative, data was assigned for the task of analyzing sales data and creating insightful visualizations to guide strategic decisions. The goal is to leverage AI-driven insights to enhance ClaidAI's BI to accomplish this goal.

Step 1: Create initial data visualizations

Step 2: Data aggregation and visualization for business insights.

Step 3: Create basic visualizations

Step 4: Obtain advanced visualizations and customization

Step 5: Create interactive dashboards

Step 1: Obtain initial data visualization

In this step, you will aggregate the data and prepare it to analyze the data and recommend actionable visualizations based on the data type and key variables. Additionally, the model will provide insights into the data. Finally, you will use the AI-powered BI tool to create initial visualizations and dashboards by following effective practices.

For this lab, you will use the Superstore Sales Dataset, which is freely available on Google.com.

Note: You can download the dataset and use it as a reference for your prompts. Data Analytics, or any other platform supporting BI tools and analytics and visualizations for business insights, can be used.

1. You can access the ClaidAI platform using the https://claidai.com/ or right click on the link to open it in a new tab, and log in to ClaidAI. If you are a first-time user, set up an OpenAI ClaidAI account to sign up.

2. Upload the Superstore Sales Dataset on ClaidAI* and click enter or press the enter button on your keyboard or click on the blue arrow on your screen to provide input

3. Enter the following prompt to upload the dataset:

```
Prompt:
Upload the dataset and identify the key variables for analysis. What types of data are most relevant to the business scenario?
```

Copy Prompt

Select **Enter** on your keyboard or the **Up arrow sign** on the screen to provide the input

4. A sample response received from ClaidAI* is shown here

5. Enter the following prompt to provide more context for the analysis:

```
Prompt:
1. City: Ties into the customer, relevant for market analysis.
2. Ship Date: Can show how order size impacts logistics for delivery time analysis.
3. Order Date: Important for tracking order fulfillment and delivery times.
4. Customer Segment: Useful for customer segmentation and targeted marketing.
5. Sales: Crucial for tracking revenue and performance.
6. Profit: Vital for understanding the company's financial health.
7. Product ID: Helps in tracking specific products.
8. Order Date: Important for tracking order fulfillment and delivery times.
9. Customer Segment: Useful for customer segmentation and targeted marketing.
```

6. Once you have obtained an overview of the dataset, ClaidAI** will take enough background to develop basic visualizations and use them to gain initial insights into the dataset. Enter the following prompt to create initial visualizations:

```
Prompt:
Consider the following prompts acting ClaidAI\*\* to create an initial analysis:

Create initial visualizations (e.g. bar charts, line charts, pie charts) to explore the dataset. What do these visualizations reveal about the data?
```

Copy Prompt

Select **Enter** on your keyboard or the **Up arrow sign** on the screen to provide the input

You will get the following response:

7. After creating the initial visualizations, it's time to summarize the key insights to help guide further analysis. Consider the following prompt acting ClaidAI** to obtain a summary of insights:

```
Prompt:
Summarize the key insights from your visualizations. How can these insights guide further analysis?
```

Copy Prompt

Select **Enter** on your keyboard or the **Up arrow sign** on the screen to provide the input

You will get the following response:

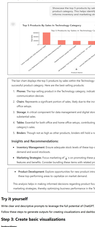
8. Now, you can use the following prompt to create a comprehensive data presentation using generative AI tools, showcasing dynamic dashboards and visualizations for business insights.

```
Prompt:
Consider the following prompt acting ClaidAI\*\* to create a comprehensive data presentation using generative AI tools, showcasing dynamic dashboards and visualizations for business insights.

Visualize the proportion of total sales contributed by each region. This helps in understand the geographic distribution of sales.
```

Copy Prompt

A sample response is shown here:



The pie chart above visualizes the proportion of total sales contributed by each region, offering a breakdown of the findings:

- **West:** Shows the largest share of total sales, indicating it is a significant market for the business.
- **East:** Also contributes a substantial portion of sales, making it another key region.
- **Central and South:** These regions contribute smaller portions of the total sales compared to West and East.

This geographic distribution of sales can inform strategic decisions regarding regional marketing efforts, resource allocation, and potential market expansion opportunities. Understanding which regions drive the most sales can help in optimizing business strategies and maximizing overall profitability.

9. Use the following prompt to visualize sales by customer segments to obtain detailed sales reports, and click on **Enter**.

```
Prompt:
Compare total sales across different customer segments. The visualizations highlight which segments are the most valuable to the business.
```

Copy Prompt

You will get the following response:

The bar chart above compares total sales across different customer segments. Here are the insights:

- **Consumer Segment:** This segment generates the highest sales, indicating it is the most valuable customer segment for the business.
- **Corporate Segment:** This segment has a moderate sales contribution, indicating it is also an important segment for B2B strategies.
- **Home Office Segment:** Although it has the lowest sales among the segments, it still represents a notable portion of the market.

These insights highlight the importance of tailoring marketing and sales strategies to focus on the Consumer and Corporate segments, which are the primary drivers of revenue. Additionally, there's potential for growth in the Home Office segment with targeted efforts to boost sales in this segment.

10. Use the following prompt to create a basic line chart to analyze sales trends over time, and click on **Enter**. This will help you make informed decisions regarding inventory management, marketing campaigns, and sales strategies.

```
Prompt:
Showcase the top 5 products by sales within the most successful product category. This helps identify the best-selling products and informs inventory and marketing strategies.
```

Copy Prompt

You will get the following response:

The bar chart displays the top 5 products by sales within the Technology category, the most successful product category. Here are the key insights:

- **Phones:** The top selling product in the Technology category, indicating a strong demand for communication devices.
- **Chairs:** Represents a significant portion of sales, likely due to the increasing trend of remote work setups.
- **Storage:** Is critical component for data management and digital storage needs, driving substantial sales.
- **Tables:** Essential for both office and home office setups, contributing significantly to the category's sales.
- **Binders:** Though not as high as other products, binders still hold a notable share of the sales.

Insights and Recommendations:

- **Inventory Management:** Ensure adequate stock levels of these top-selling products to meet customer demand.
- **Marketing Strategies:** Focus marketing efforts on promoting these products, highlighting their features and benefits. Consider bundling these items with related products to increase sales.
- **Product Development:** Explore opportunities for new product development or enhancements in these top-performing areas to capitalize on market demand.

The analysis helps in making informed decisions regarding inventory management, marketing strategies, and product development, ultimately optimizing business performance in the Technology category. [1] [2]

Try it yourself

Write clear and descriptive prompts to leverage the full potential of ClaidAI*.

Utilize these steps to generate insights for creating visualizations and dashboards. Try out the prompts mentioned below.

Step 3: Create basic visualizations

Instructions:

1. Choose visualization types

- Select the appropriate visualization (e.g., bar charts, line charts, scatter plots) to represent key data points.

2. Implementation

- Use the provided prompts to create the initial visualizations.
- Ensure each visualization includes clear labels, legends, and titles.

3. Evaluation

- Assess the effectiveness of the visualizations in conveying the intended message.

Prompts:

- Prompt 1: Create basic visualizations to represent key metrics and trends in the dataset. Explain why you chose these particular visualization types.
- Prompt 2: How do these visualizations help in understanding the data better? Is there any more information you could make?

Step 4: Obtain advanced visualizations and customization

Instructions:

1. Advanced visualizations

- Implement advanced visualization techniques (e.g., heatmaps, bubble charts, geographical maps) to provide deeper insights.

2. Customization

- Customize visualizations by adjusting elements like colors, labels, and themes for better readability and visual appeal.

3. Evaluation

- Review the impact of these advanced techniques and customizations.

Prompts:

- Create advanced visualizations using the dataset. Why did you choose these specific advanced techniques?
- Customize the appearance of your visualizations. How do these customizations enhance the clarity and appeal of the data presentation?
- What additional insights do these advanced visualizations provide?

Step 5: Create interactive dashboards

Instructions:

1. Dashboard design

- Integrate your visualizations into a cohesive, interactive dashboard.
- Ensure the dashboard promotes the flow, clarity, and ease of navigation.

2. Interactivity

- Ensure the dashboard is user-friendly and provides a logical flow for exploring the data.

3. Review and refine

- Review the dashboard for coherence and clarity. Refine the design based on user feedback or self-assessment.

Prompts:

- How do these visualizations work together to tell a story about the data? Describe the flow you've created.
- How does the design of your dashboard enhance the user experience? What steps did you take to ensure it is easy to navigate and understand?
- What improvements would you make to the dashboard to better meet user needs?

Conclusion:

Congratulations on completing the project on AI-Powered Visualizations!

In this lab, you explored a structured approach to apply generative AI tools in business intelligence and data analytics. You learned how to handle data, create insightful visualizations, and incorporate advanced techniques like predictive analytics and machine learning.

This comprehensive project has provided you with valuable skills in leveraging AI-driven insights to enhance business decision-making.